

CHAPTER ONE

How to Use This Book

The three sports of mountain running, skimo, and ski mountaineering (skimo and ski mountaineering differ in the tempo and hence intensity of uphill skinning due to the weight of the equipment needed for ski mountaineering) are complementary in most of the demands made on the body. Many skiers run for training and increasingly, runners are finding their way to skiing as a fun and valuable addition to their training. All three of us have used the complementary nature of these sports for many years. It was only natural to combine the topics of training for each into one book. Where the three sports' training converges, we try as much as possible to combine them in the discussion. Where differences cause the training to diverge, we make clear why and how to apply different training techniques. Despite the heavy emphasis on competition, the methods explained in this book work equally well for the recreational level or noncompetitive mountain athlete too. Your body faces the same physiological limitations regardless of whether you have a race number on or are only interested in the more solitary challenges of a big day in the mountains. We think both sets of readers can learn from these pages.

We wrote this book to share three lifetimes of knowledge and experience on the science and methodology of endurance training. But we need to start with a disclaimer: Collectively, the three of us have made all the training mistakes and a few of them more than once. We know from painful firsthand experience just as much about how *not* to train effectively as we have learned about how to train correctly. There are many more ways to train wrong than right. While learning from your own mistakes is the best teacher, it is painful and perhaps more importantly, a terrible waste of time. By steering you around the pitfalls

and danger areas, we hope to save you wasted days and weeks, to say nothing of the disappointment and even heartbreak of ineffective training. Over the accumulated decades that we have been training and coaching, we have been able to refine our approach and want to share this with you.

Physical training is often presented in a very formulaic, cookbook approach: follow the recipe and your fitness cake will turn out perfectly. The reality is that we are very complex organisms and identical recipes will not work for everyone. Of course, all humans share the same physiological systems but the responses (both in quantity and in kind) to identical training inputs can vary a great deal among any group. Said another way: If a coach has ten athletes perform the identical training session and measures certain physiologic systems, he will see ten different responses. While it is important to give general guidelines and training prescriptions for athletes to follow, unless they have some idea of their individual response, they will be flying blind.

As a consequence, this book is not a cookbook. You will not find a recipe to make one cake. What you will find is an explanation of some of the underlying systems that account for the variation among athletes. You'll learn how to determine what your individual responses to training are and what they mean. In total, our goal is to convey a set of tools that will allow you to operate within a fairly narrow range of training options but be able to individualize the minutiae of training so that it yields the best possible results for you.

The study of any subject is best managed within a framework. We hope to establish that framework within which to dissect what is a complex topic. Even though none of this material is difficult to grasp, there are a lot of topics covered and there's no way one could absorb all the information presented here in one reading. It is our hope that you will use this book as a reference manual to guide your training over many seasons as you apply what we have spent decades learning. No single part of the book is meant to stand alone. Each section is supported by the preceding ones and as such supports the subsequent parts.

We start in Section One with a look at The Physiological Basis of Endurance Training. Despite the limitations of idealized scientific studies, they can still provide us with a better understanding of the *whys* even if they can't help us so much with the *whats* or *hows*. These *whats* and *hows* are where endurance training becomes more art than science. And like any subject, it can be learned only by study. This section provides the theoretical framework for what is to come.

In Section Two, The Methodology of Endurance Training, we identify and explain numerous concepts that, if understood and applied correctly, will make you a better athlete or coach. These are the tools in your toolbox. Knowing which tool is appropriate for each task is what makes the artisan a master. The range of topics here is as broad as it is deep and this is the area of the book that will likely be the one you go back to over and over again to refresh your knowledge.



As fall transitions to winter, Davide Stofie does a ski-bounding workout in Passo Oclini in South Tyrol, Italy. Photo: Federico Modica

Section Three is devoted to Strength Training for the Uphill Athlete. We demystify strength in terms most applicable to endurance athletes (as opposed to weight lifters).

By the time we get to Section Four we will be focusing on How to Train. Here we talk about ways to categorize, design, and structure workouts as well as how to mix them depending on what you want to achieve. All the previous information will be brought into play as we show you how to progress from very basic, but essential aerobic-capacity building to the speed endurance required for your event. And finally, we offer several example plans you can use for guidance or apply directly to your own training.

